Jinhan Kim

AI and Software Engineering

Curriculum Vitae / NOV 2026

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01 APPOINTMENTS

Postdoctoral Researcher — **Sep 2023 – Present**
Università della Svizzera italiana (USI), Lugano, Switzerland / Advisor: Prof. Paolo Tonella / Funding: the European Horizon project Sec4AI4Sec and the NRF Korea Postdoctoral Fellowship Program 2024 / Research on testing and assurance for deep-learning and RAG systems.

Postdoctoral Researcher — **Mar 2023 – Aug 2023**
KAIST, Daejeon, South Korea / Advisor: Prof. Shin Yoo / Postdoctoral research at the COINSE lab.

Visiting Ph.D. Student — **Aug 2022 – Sep 2022**
Università della Svizzera italiana (USI), Lugano, Switzerland / Advisor: Prof. Paolo Tonella / Funding: Young Researchers' Exchange Programme between South Korea and Switzerland 2022 / Research on mutation-based deep-learning system testing and deep-learning program repair.

Research Intern, Users & Information Lab — **Sep 2015 – Dec 2015**
KAIST, Daejeon, South Korea / Advisor: Prof. Alice Oh / Designed and deployed EliceQ, a social back-channel application for anonymous classroom questions.

Research Intern, NC Lab — **Mar 2015 – Aug 2015**
KAIST, Daejeon, South Korea / Advisor: Prof. Junehwa Song / Developed BeUpright and supported a two-week human study on relational-norm interventions for posture change.

02 EDUCATION

Ph.D. in Computer Science — **Mar 2017 – Feb 2023**
KAIST, Daejeon, South Korea
Integrated M.S./Ph.D. programme. • Advisor: Prof. Shin Yoo.

B.S. in Computer Science — **Mar 2012 – Feb 2017**
KAIST, Daejeon, South Korea

03 PH.D. SUPERVISION

Masoud Jamshidiyan Tehrani — **Feb 2024 – Present**
Ph.D. student at USI; co-supervised with Prof. Paolo Tonella

Samuele Pasini — **Oct 2023 – Present**
Ph.D. student at USI; co-supervised with Prof. Paolo Tonella

04 AWARDS, HONORS & PATENT

2023 **Best Paper Award** 18th International Workshop on Mutation Analysis

2023 **KAIST Ph.D. Dissertation Award** College of Engineering, KAIST

2020 **NAVER Ph.D. Fellowship Award** School of Computing, KAIST

2021 **Korea Patent No. 1020190104591** *Method for Evaluating Test Fitness of Input Data for Neural Network and Apparatus Thereof*; published August 9, 2021.

05 PUBLICATIONS

Selected papers appear first; Jinhan Kim appears in navy. All are main-track unless noted.

Google Scholar | 12 Aug 2026 | Citations **1,067** | h-index **12** | i10-index **14**

SELECTED PUBLICATIONS

[1] ICSE · 2027 **Rethinking Mutation Testing When the Natural Language Specification Is the Program**
J. Kim and P. Tonella • *Proceedings of the 49th IEEE/ACM International Conference on Software Engineering*.

[2] EMSE · 2026 **Real Faults in Deep Learning Fault Benchmarks: How Real Are They?**
G. Jahangirova, N. Humbatova, **J. Kim**, S. Yoo, and P. Tonella • *Empirical Software Engineering*.

[3] ASE · 2026 **When Knowledge Changes: Metamorphic Testing of RAG Systems with Mutations**
J. Kim, S. Pasini, and P. Tonella • *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering*.

[4] ISSTA · 2026 **Testing Retrieval-Augmented Generation Systems with Chunk Coverage**
J. Kim, S. Pasini, and P. Tonella • *Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis*.

[5] ICSE · 2026 **Revisiting "Revisiting Neuron Coverage for DNN Testing: A Layer-Wise and Distribution-Aware Criterion": A Critical Review and Implications on DNN Coverage Testing**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella • *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering*.

[6] TOSEM · 2025 **A Taxonomy of System-Level Attacks on Deep Learning Models in Autonomous Vehicles**
M. J. Tehrani, **J. Kim**, R. Z. L. Foulefack, A. Marchetto, and P. Tonella • *ACM Transactions on Software Engineering and Methodology*.
∗ *Journal-first presentation at ICSE 2026*

[7] EMSE · 2025 **Evaluating and Improving the Robustness of Security Attack Detectors Generated by LLMs**
S. Pasini, **J. Kim**, T. Aiello, R. Lozoya, A. Sabetta, and P. Tonella • *Empirical Software Engineering*.
∗ *Journal-first presentation at ICSE 2026*

[8] EMSE · 2025 **An Empirical Study of Fault Localisation Techniques for Deep Neural Networks**
N. Humbatova, **J. Kim**, G. Jahangirova, S. Yoo, and P. Tonella • *Empirical Software Engineering*, 30(5), p. 124.
∗ *Journal-first presentation at ICST 2026*

[9] TOSEM · 2023 **Evaluating Surprise Adequacy for Deep Learning System Testing**
J. Kim, R. Feldt, and S. Yoo • *ACM Transactions on Software Engineering and Methodology*, 32(2), pp. 1–29.

[10] ICST · 2023 **Repairing DNN Architecture: Are We There Yet?**
J. Kim, N. Humbatova, G. Jahangirova, P. Tonella, and S. Yoo • *Proceedings of the 16th IEEE International Conference on Software Testing, Verification and Validation*, pp. 234–245.

[11] TOSEM · 2022 **Predictive Mutation Analysis via Natural Language Channel in Source Code**
J. Kim, J. Jeon, S. Hong, and S. Yoo • *ACM Transactions on Software Engineering and Methodology*, 31(4), pp. 1–27.
∗ *Journal-first presentation at ICSE 2023*

[12] ICSE · 2019 **Guiding Deep Learning System Testing Using Surprise Adequacy**
J. Kim, R. Feldt, and S. Yoo • *Proceedings of the 41st IEEE/ACM International Conference on Software Engineering*, pp. 1039–1049.

OTHER JOURNAL ARTICLES

[1] **JSS · 2026** **Cross-Site Scripting Adversarial Attacks Based on Deep Reinforcement Learning: Evaluation and Extension Study**
S. Pasini, G. Maragliano, **J. Kim**, and P. Tonella • *Journal of Systems and Software*.

[2] **IST · 2023** **Learning Test-Mutant Relationship for Accurate Fault Localisation**
J. Kim, G. An, R. Feldt, and S. Yoo • *Information and Software Technology*, 162, p. 107272.

OTHER CONFERENCES & WORKSHOPS

[1] **TRUST · 2026** **Dynamic Deception: When Pedestrians Team Up to Fool Autonomous Cars**
M. Tehrani, M. Gabriel, **J. Kim**, and P. Tonella • *Proceedings of the 1st International Workshop on Trustworthy and Responsible aUtonomous SysTems*.

[2] **FSE Demo · 2025** **PCLA: A Framework for Testing Autonomous Agents in the CARLA Simulator**
M. Tehrani, **J. Kim**, and P. Tonella • *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, pp. 1040–1044.

[3] **Mutation · 2023** **The Inversive Relationship Between Bugs and Patches: An Empirical Study**
J. Kim, J. Park, and S. Yoo • *Proceedings of the 18th International Workshop on Mutation Analysis*, pp. 314–323.

[4] **ICST Industry · 2022** **Repairing Fragile GUI Test Cases Using Word and Layout Embedding**
J. Yoon, S. Chung, K. Shin, **J. Kim**, S. Hong, and S. Yoo • *Proceedings of the 15th IEEE International Conference on Software Testing, Verification and Validation*.

[5] **ISSRE · 2021** **Ahead of Time Mutation Based Fault Localisation Using Statistical Inference**
J. Kim, G. An, R. Feldt, and S. Yoo • *Proceedings of the 32nd International Symposium on Software Reliability Engineering*, pp. 253–263.

[6] **FSE Industry · 2020** **Reducing DNN Labelling Cost Using Surprise Adequacy: An Industrial Case Study for Autonomous Driving**
J. Kim, J. Ju, R. Feldt, and S. Yoo • *Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, pp. 1466–1476.

[7] **GI · 2018** **Comparing Line and AST Granularity Level for Program Repair Using PyGGI**
G. An, **J. Kim**, and S. Yoo • *Proceedings of the 4th Genetic Improvement Workshop*, pp. 19–26.

[8] **L@S · 2018** **Elicast: Embedding Interactive Exercises in Instructional Programming Screencasts**
J. Park, Y. H. Park, **J. Kim**, J. Cha, S. Kim, and A. Oh • *Proceedings of the Fifth Annual ACM Conference on Learning at Scale*, pp. 1–10.

[9] **SSBSE · 2018** **Learning Without Peeking: Secure Multi-Party Computation Genetic Programming**
J. Kim, M. G. Epitropakis, and S. Yoo • *Proceedings of the 10th International Symposium on Search Based Software Engineering*, pp. 246–261.

[10] **KCSE · 2017** **PyGGI: Python General framework for Genetic Improvement**
G. An, **J. Kim**, S. Lee, and S. Yoo • *Proceedings of Korea Software Congress*, pp. 536–538.

[11] **SSBSE Short Papers · 2017** **GPGPGPU: Evaluation of Parallelisation of Genetic Programming Using GPGPU**
J. Kim, J. Kim, and S. Yoo • *Proceedings of the 9th International Symposium on Search Based Software Engineering*, pp. 137–142.

[12] **CHI · 2016** **BeUpright: Posture Correction Using Relational Norm Intervention**
J. Shin, B. Kang, T. Park, J. Huh, **J. Kim**, and J. Song • *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pp. 6040–6052.

[13] **SenSys Demo · 2015** **Demo: Posture Correction Using Smartphone-Based Relational Intervention Model**
J. Shin, B. Kang, **J. Kim**, J. Huh, J. Song, and T. Park • *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems*, pp. 495–496.

PREPRINTS & MANUSCRIPTS

[1] **arXiv · 2026** **Detecting Trojaned DNNs via Spectral Regression Analysis**
S. Pasini, **J. Kim**, and P. Tonella • *arXiv preprint*.

[2] **arXiv · 2025** **Fault Localisation and Repair for DL Systems: An Empirical Study with LLMs**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella • *arXiv preprint*.
∗ *Under revision at ACM Transactions on Software Engineering and Methodology*

[3] **arXiv · 2025** **MuFF: Stable and Sensitive Post-training Mutation Testing for Deep Learning**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella • *arXiv preprint*.

[4] **arXiv · 2025** **Mutation-Guided Test Generation for DNNs Using Diffusion**
S. Kim, **J. Kim**, V. Riccio, S. Yoo, and P. Tonella • *arXiv preprint*.

06 ACADEMIC SERVICE

Organising Committees

2027 ICST, Posters Track Co-chair
2026 ICST, Testing Tools and Data Showcase Track Co-chair; SSBSE, Publicity Co-chair; DeepTest, Workshop Co-organiser; SBFT, Workshop Co-organiser
2025 DeepTest, Workshop Co-organiser

Program Committees

2027 ICSE, Research Track
2026 ASE, Research Track; ESEM, Research Track; ICST, Education Track; Mutation; RASE
2025 ISSTA, Research Track; ASE, NIER Track and Student Research Competition; SSBSE, Challenge Track; SBFT; Mutation; DeMeSSAI
2024 SCAM, Research Track; ICSE, Demonstrations Track; ASE, Demonstrations and NIER Tracks; DeepTest; Mutation
2023 Mutation; ASE, NIER Track; ICSME, Artifact Evaluation Track and ROSE Festival
2022 Mutation; ICST, Poster Track; ICSME, Registered Reports Track, Artifact Evaluation Track, and ROSE Festival
2021 Mutation; ICSME, Artifact Evaluation Track
2020 Mutation

Journal Boards & Reviewing

2026 Reviewer: Empirical Software Engineering
2025 TOSEM Board of Distinguished Reviewers; reviewer for TSE, TOSEM, EMSE, STVR, and IEEE Computer
2024 TOSEM Board of Distinguished Reviewers; reviewer for TSE, TOSEM, EMSE, JSEP, and JSS
2023 Reviewer for TSE, TOSEM, EMSE, and JSEP
2022 Reviewer for TOSEM and JSEP
2021 Reviewer for TOSEM and STVR
2020 Reviewer for IST and JSS

07 INVITED TALKS

2026 **Mutation Workshop, ICST 2026, Daejeon, South Korea** / Panellist.

2026 **Software Institute Seminar, USI, Lugano, Switzerland** / *Testing RAG Systems: Coverage-Guided Adequacy and Metamorphic Oracles*.

2026 **CHOOSE Forum, Bern, Switzerland** / *Breaking DNNs: Mutation Testing for Deep Neural Networks.*

2024 **Software Institute Seminar, USI, Lugano, Switzerland** / *When Simple is Better than Complex: Coverage and Mutation for DL Testing.*

2023 **Korea Conference on Software Engineering, Pyeongchang, South Korea** / Invited paper presentation: *Predictive Mutation Analysis via Natural Language Channel in Source Code.*

2022 **Korea Conference on Software Engineering, Pyeongchang, South Korea** / Invited paper presentation: *Ahead of Time Mutation Based Fault Localisation using Statistical Inference.*

2019 **Korea Software Congress, Pyeongchang, South Korea** / Invited paper presentation: *Guiding Deep Learning System Testing Using Surprise Adequacy.*

08 TEACHING

KAIST CS101: Introduction to Programming – Teaching Assistant, Fall 2016, Spring 2020, Fall 2020, Spring 2021.
KAIST CS453: Automated Software Testing – Teaching Assistant, Spring 2018, Spring 2019.
KAIST CS454: AI Based Software Engineering – Teaching Assistant, Fall 2021.
KAIST CS489: Computer Ethics and Social Issues – Teaching Assistant, Fall 2019.

09 REFERENCES

Prof. Shin Yoo	Full Professor, KAIST • Daejeon, South Korea	shin.yoo@kaist.ac.kr
Prof. Paolo Tonella	Full Professor, USI • Lugano, Switzerland	paolo.tonella@usi.ch
Prof. Robert Feldt	Full Professor, Chalmers • Gothenburg, Sweden	robert.feldt@chalmers.se